

Hongbo Yu

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EDUCATION

Rice University

Master of Computer Science

Houston, TX

Aug. 2026 – Dec. 2027 (Expected)

Shenzhen University

B.E. in Computer Science and Technology

Shenzhen, China

Sep. 2021 – Jul. 2025

EXPERIENCE

Agent Development Intern

Autoagents

Mar. 2026 – Jul. 2026

Beijing, China

- Refactored linear workflows into a **LangGraph multi-agent architecture** with explicit state transitions; added **Redis-backed checkpointing** to persist execution state and resume asynchronous agent sessions.
- Improved agent tool-calling reliability with **Pydantic structured-output validation**, automatic retries, and self-correction loops, recovering from malformed model responses instead of failing the entire workflow.
- Built an agent evaluation workflow with **LangSmith tracing** and golden test cases to analyze decision trajectories and compare prompt changes; integrated regression testing into CI/CD to catch agent behavior regressions.
- Deployed agent services and vector databases with **Docker, Kubernetes, and Helm**; configured autoscaling and health probes to support production workloads while maintaining **99.9% service availability**.

Backend Development Intern

Aqumon (Hongliang Research)

Mar. 2025 – Jul. 2025

Shenzhen, China

- Designed a **JWT + Redis** authentication system with access and refresh tokens, sliding expiration, and claims-based validation for multi-tenant APIs, enabling gateway verification without repeated database lookups.
- Integrated **Stripe** with **RocketMQ** for asynchronous payments; implemented Redis idempotency keys, database constraints, and explicit order-state transitions to prevent duplicate transactions.
- Optimized **MySQL** queries and redesigned composite and covering indexes, reducing core API **p99 latency from 500ms to under 200ms**.

PROJECTS

FitCore – AI-Powered Fitness Platform | *Python, FastAPI, pgvector, LoRA*

Jan. 2026 – Present

Demo: fitcore-web-eight.vercel.app | **GitHub:** github.com/G1lollipop/fitcore

- Built a **hybrid RAG pipeline** for fitness and nutrition queries, combining Gemini vector embeddings with **BM25** keyword retrieval and a **bge-reranker-base CrossEncoder** to rerank candidate context.
- Implemented adaptive top-*k* retrieval and confidence-based abstention gates, adjusting the amount of retrieved context and filtering low-confidence results before passing fitness and nutrition context to the language model.
- Fine-tuned a lightweight language model with **LoRA/SFT** to extract calories and macronutrients from free-form food descriptions; enforced schema-constrained JSON outputs with inference latency under **150ms**.
- Built and deployed the application with **FastAPI, Next.js, and Supabase**; delivered AI chat through **Server-Sent Events (SSE)** and optimistic UI updates, keeping time-to-first-token under **1 second**.

SKILLS

Languages: Python, Java, C++, Go, SQL, TypeScript

AI & LLMs: PyTorch, LangGraph, LangChain, RAG, LoRA/SFT, Tool Calling

Retrieval & Data: pgvector, Milvus, ChromaDB, Elasticsearch, Redis, MySQL

Evaluation & Serving: LangSmith, Pydantic, FastAPI, Supabase

Engineering: Docker, Kubernetes, Helm, Linux, Git, Spring Boot, Next.js